Bitcoin RL Trading with Directional Change Events + PPO

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Agenda

1. Motivation & Problem Setup
2. Market Opportunities
3. Deep Reinforcement Learning Overview
4. Directional Change Framework
5. Our Model
   - PPO Agent Design
   - Reward Function
   - Evaluation
   - Results Summary
   - Future Works

Team

Fred Montero Arteaga

Strategy | Operations & Analytics | Engineering

- Experiences: *Oliver Wyman, Boeing, Spectrum Plastics Group, LiveFree Personal Protection*
- Entrepreneurship Through Acquisition (ETA)
- New dad!

Jim Long

Equity Research | Quant Investing

- Experiences: *Adage Capital Management , 4S Bay Partners, KeyBanc Capital Markets, KPMG*
- 1st place T. Rowe Price impact investing competition; CFA Institute Investing Simulation Challenge global finalist
- Chartered Financial Analyst, Chartered Market Technician

Project Support

Advisor: Prof. Nikolay Kadochnikov

TA: Josh Rauvola

Bitcoin & Crypto Market Economic Significance

Market Scale | Liquidity & Trading Activity | Market Structure | Growth Drivers | Implications | Structural Tailwinds

- Global crypto market cap: $1–3+ trillion (cycle-dependent)
- Bitcoin represents 40–55% of total market value
- Growing institutional access via ETFs, custody, and derivatives

- Bitcoin spot volume: $10–30+ billion daily
- Total crypto volume: $50–100+ billion daily in volatile periods
- 24/7 global trading with no market close

- Significant turnover in futures and perpetual swaps
- Open interest often in the tens of billions of dollars
- Leverage amplifies volatility and regime shifts

- Rising institutional adoption and treasury allocations
- Expansion of regulated trading venues globally
- Increasing algorithmic and quantitative participation

- Scale and liquidity make systematic strategies economically meaningful
- High volatility creates a rich sequential learning environment
- Structural evolution supports long-term research relevance

- Institutional adoption with treasury allocations increasing crypto exposure
- Exchanges are securing licenses and compliance frameworks
- Embedding into broader financial infrastructure

Policy Bitcoin is not a niche asset class, it is a trillion-dollar, globally traded market whose structural volatility makes adaptive decision systems economically valuable today and in the future.

Motivation & Problem Setup

Architecting an autonomous trading agent for cryptocurrency markets using Deep Reinforcement Learning.

Market Dynamics

Bitcoin's 24/7 liquidity and intense volatility present substantial alpha. Traditional heuristic strategies often fail to dynamically adapt to its non-stationary microstructural shifts.

Algorithmic Edge

Supervised models merely predict prices. Trading requires sequential, risk-adjusted decisions under uncertainty. RL directly optimizes for cumulative profitability and drawdown control.

MDP Environment

Modeled as a Partially Observable Markov Decision Process (POMDP). The state space observes normalized OHLCV metrics, technical indicators, and active balance.

Action & Reward

The agent outputs an optimal allocation vector . The objective maximizes the step-wise log return, explicitly penalized for excessive turnover to mimic real costs.

Market Challenges & Opportunities

Strategic Opportunities

Regime Dynamics in Crypto Markets
Bitcoin exhibits recurring directional moves and momentum phases.

RL Advantage
RL directly optimizes long-term cumulative returns rather than short-term price forecasts.

Event-Based Sampling
Event-driven frameworks reduce noise and align decisions with meaningful structural shifts.

Market Challenges

Extreme Volatility
Frequent regime shifts and non-stationary return distributions.

Limitations of Traditional Models
Supervised models focus on predicting price, not optimizing sequential trading decisions.

Sequential Decision Complexity
Trading performance depends on dynamic capital allocation under uncertainty.

Deep Reinforcement Learning Overview

Environment

Financial markets can be modeled as partially observable systems.
State variables may include price information (e.g., OHLCV), technical indicators, volatility measures, and portfolio exposure.
Observations can be sampled either in fixed time intervals or event-driven frameworks.

Agent

In reinforcement learning, an agent interacts with the market by selecting actions based on observed states. Common approaches use actor critic architectures:
The policy (actor) proposes actions
The value function (critic) evaluates expected future rewards
Typical actions: Buy, Hold, or Sell.

Objective

The goal is to maximize cumulative reward, often defined in terms of risk-adjusted returns.
Reward functions typically incorporate:
Profit & Loss (P&L)
Transaction costs
Turnover penalties
Drawdown or risk constraints

- Policy regularization techniques (e.g., PPO clipping) help stabilize learning in volatile reward landscapes.

- Event-driven decision frameworks allow agents to act at structurally meaningful moments rather than fixed time intervals.

RL Policies and Agent Learning

Proximal Policy Optimization 01.

Policy-gradient methods balance exploration, stability, and risk control

- **Clipped Objective:** Constraints policy updates to avoid large, & destabilizing shifts
- **Sample Efficient:** Learns from finite trajectories while maximizing informational content
- **Robust Reward Paths:** Designed to handle stochastic, & heavy-tailed financial return d

Learning & Regularization 02.

Layered safeguards promote stable learning and reduce overfitting

- **Training Controls:** Careful tuning of KL limits and entropy to balance exploration and stability
- **Validation Controls:** RiskEval checkpoint filtering, and benchmark-aware scoring
- **Implementation Controls:** Position constraints, exposure limits, and minimum confidence thresholds

Architecture & Execution Logic 03.

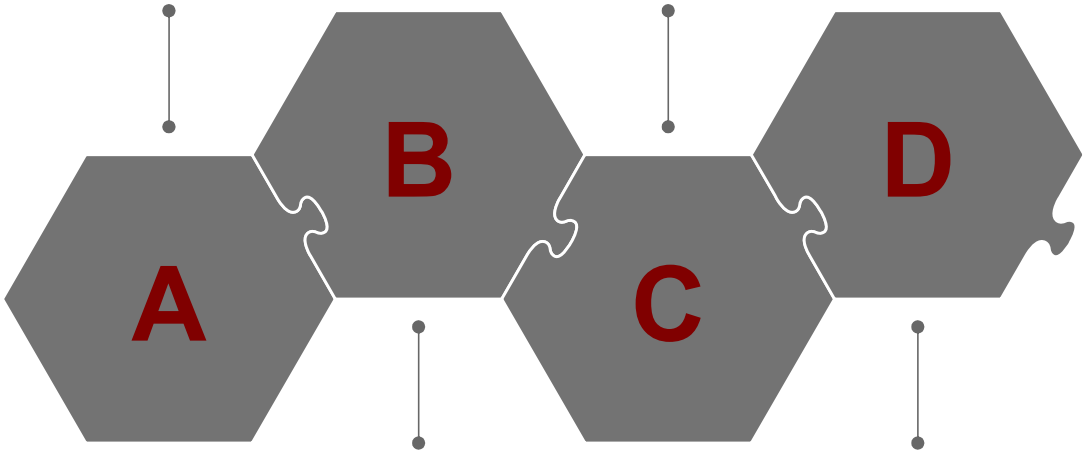
Market frictions and execution mechanics significantly impact realized performance

- **Event-Driven Triggers:** Initiated by market events rather than fixed time intervals
- **Execution Latency:** Order delays, and market impact affect realized returns
- **Portfolio Update Logic:** Positions are rebalanced, and portfolio state variables are updated after each action

Directional Change Methodology

Event-Based Sampling: Transitions from fixed time intervals to intrinsic time, sampling only when price shifts by a set threshold.

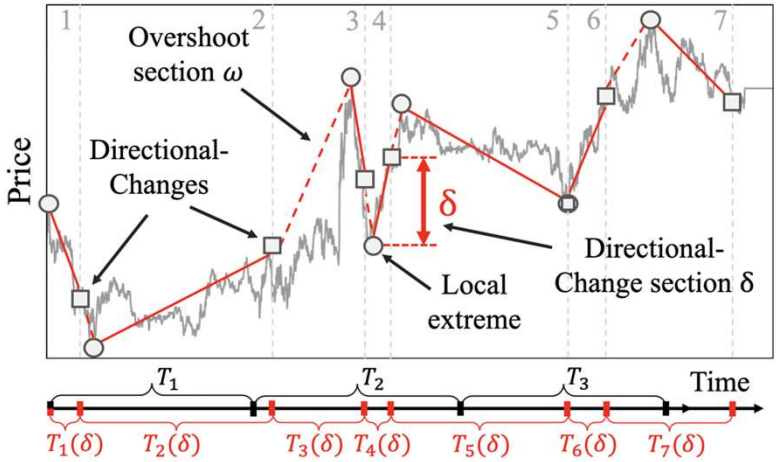
Noise Filtration: Eradicates extreme cryptocurrency micro-volatility, preventing the agent from acting on false signals.



DC & OS Phases: A Directional Change (DC) confirms a reversal. An Overshoot (OS) captures continued momentum.

State Compression: Reduces the MDP state dimension, accelerating RL convergence and improving structural generalization.

A DC event triggers when price moves by a fixed threshold (0.25%) from the last extreme. After the event, an **overshoot phase** begins until the next reversal, creating monetizable continuation opportunities.



Keywords: puzzles

Our Model

Proximal Policy Optimization (PPO) Selection

Why PPO?

Clipped objective → stable policy updates in noisy reward landscapes
Sample efficient → critical with finite DC events
Discrete action space: Buy / Hold / Sell

Regularization Stack

Clip range + KL target + entropy decay
Risk-based checkpoint filtering
Action gates (long/short/hold constraints)

Execution: $1 starting capital, full reinvestment, 2-min execution lag. Decisions trigger *only* at DC events.

Actor-Critic Architecture

Decouples trading logic. The Actor continuously outputs action projection, while the Critic evaluates the state-value to compute baseline-adjusted advantages.

Objective Clipping

Restricts the policy update ratio within a strictly defined trust region. This prevents catastrophic over-optimization to single noisy price spikes in volatile BTC markets.

Sample Efficiency

Unlike vanilla policy gradients, PPO safely reuses historical price trajectories for multiple stochastic gradient ascent epochs, maximizing data utility.

PPO Agent: State Design

First Iteration

- **Hourly to daily features**
- Returns: 1h/1d/30d/60d
- Volatility: 1h/1d/30d/60d
- Tick volume avg: 1h/1d/30d/60d
- DC event descriptors: direction, event_code, move%, move duration

Pro: easy to construct
Con: insufficient edge for high-frequency decision in minutes, some data are not stationary (eg volume)

Second Iteration

- **Added minute-level features**
- Returns: 5min/10min//15min/30min
- Volatility: 5min/10min//15min/30min
- Tick volume avg: 5min/10min//15min/30min

Pro: improved state characteristics
Con: still not enough for high-frequency decision

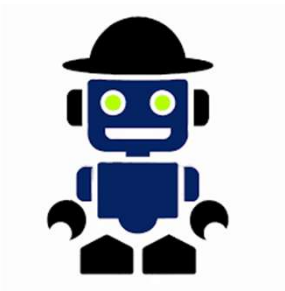
Third Iteration

- **Added technical indicators**
- Distance to exponential moving avg: 5min to 60d
- Volatility ratio: 5m/30m, 15m/1h, 30m/1d
- Relative strength: 5min to 1d
- Moving average convergence/divergence (MACD): 5min to 1d

Pro: improved understanding of states
Con: still not enough to distinguish regime change

Future works: 1) add exogenous signals, such as VIX, cross-asset signal, sentiment; 2) incorporate sequence models (LSTM/Transformer) to create predictions

PPO Agent: Actions & Environment

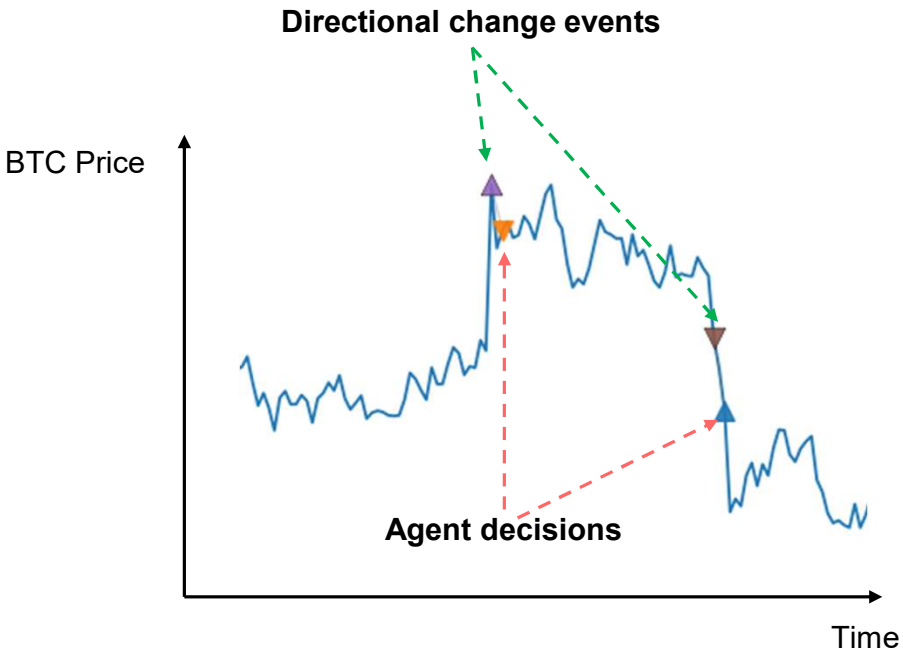


Action Space: Buy, Hold (no-exposure), Sell

Capital: $1 initial capital, and no subsequent injection

Decision point: directional change event

Latency: 2 mins lag to buffer decision-making and trade execution



Future works: 1) explore partial position sizing; 2) explore variable execution latency and impact on the agent's edge

Reward Function

Reward = Scale factor * (profit and loss – transaction cost – penalties)

100

A * absolute profit + (1-A) outperformance

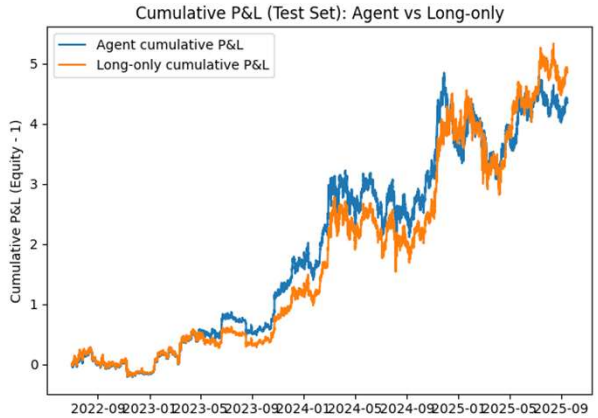
Turnover
Downside (single action)
Rolling drawdown
Hold (laziness)

=== Reward component summary ===

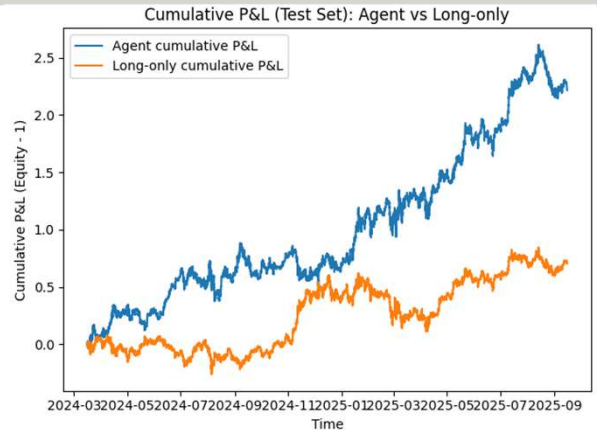
	component	mean	std	sum	abs_sum	non_na
0	reward	-0.032998	0.249186	-2.804850	13.438429	85
1	pnl	-0.000202	0.003889	-0.017153	0.211158	85
2	pnl_used	-0.000120	0.002335	-0.010219	0.127926	85
3	turnover_pen_term	0.000118	0.000473	0.010000	0.010000	85
4	shaping_total	-0.000092	0.000199	-0.007829	0.007829	85
5	roll_dd_pen_term	0.000060	0.000101	0.005119	0.005119	85
6	active_pnl	0.000002	0.000238	0.000181	0.003079	85
7	downside_pen_term	0.000032	0.000179	0.002710	0.002710	85
8	cost_term	0.000000	0.000000	0.000000	0.000000	85
9	hold_pen_term	0.000000	0.000000	0.000000	0.000000	85

Evaluation (One-time train/test, no execution lag)

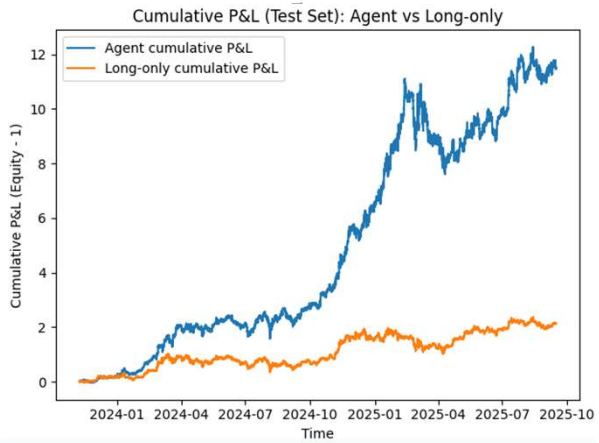
60/40 Train/Test



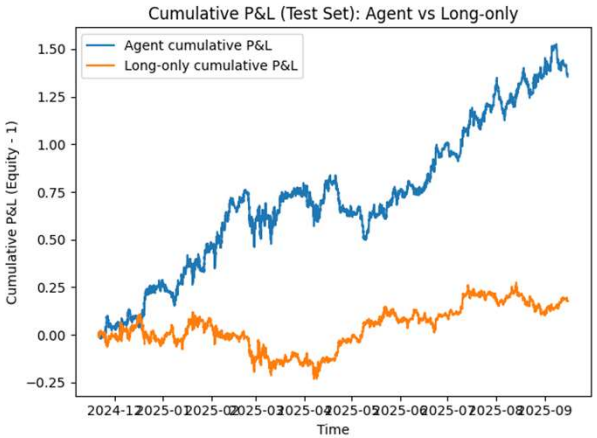
80/20 Train/Test



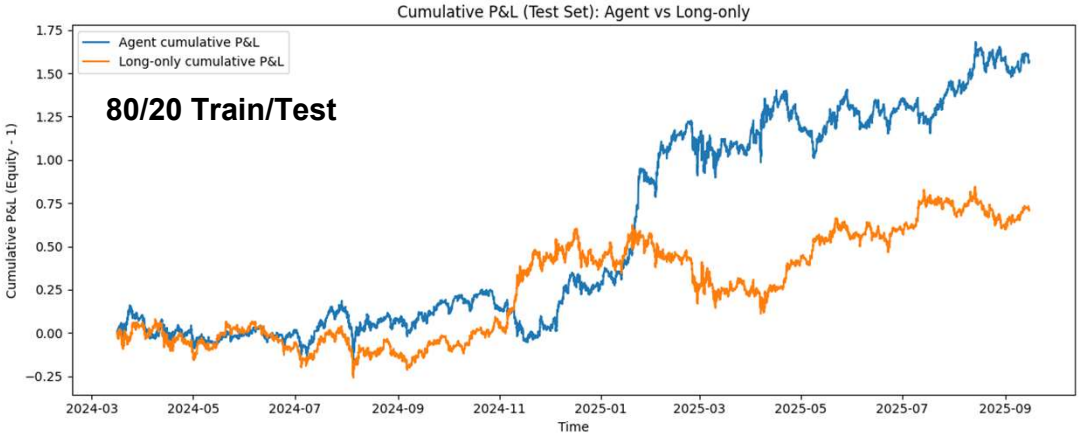
75/25 Train/Test



90/10 Train/Test



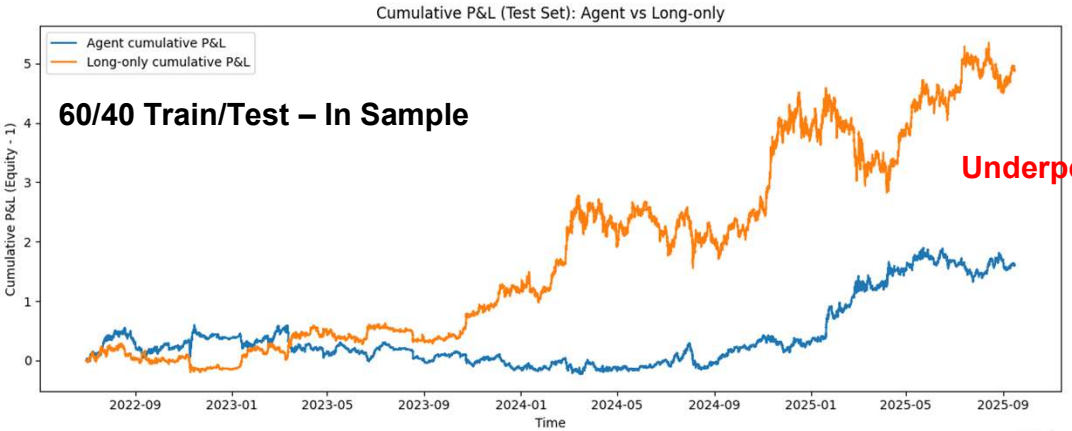
Evaluation (One-time train/test, with execution lag)



Observations:

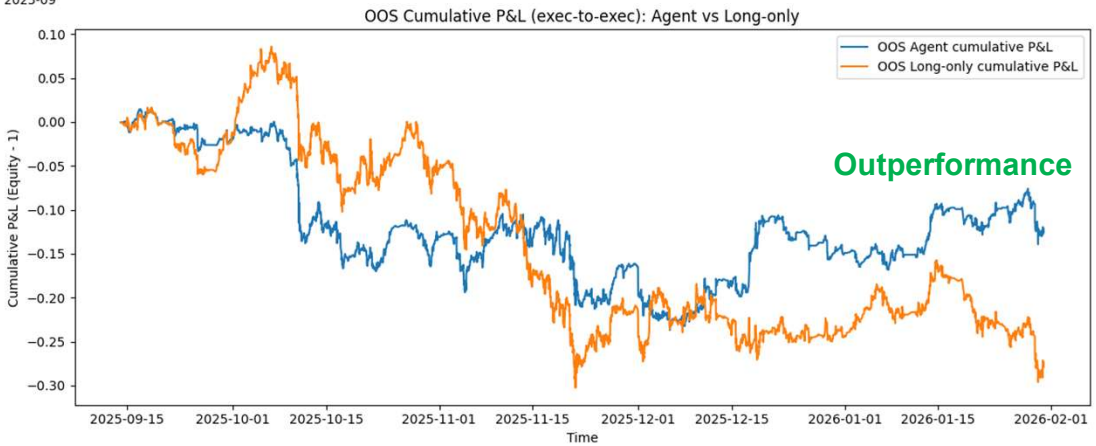
- Agent performances still vary across train/test splits.
- We expect each execution lag to require its own model optimization, while a shorter lag seems to better edge (profits).

Evaluation (One-time train/test, with execution lag)



Underperformance

60/40 Train/Test – Out-of-Sample



Outperformance

Observations:

- Agent could underperform in-sample, but outperform out-of-sample.
- Generalization for unseen data requires further model refinement.

Results Summary

Performance Observations

Performance varies notably across static train/test splits. The agent doesn't consistently outperform in every configuration, which motivates the more adaptive training approaches we're pursuing in future work.

Execution Lag Impact

A standard **2-minute lag** noticeably reduces the trading edge.
Shorter execution lags drastically improve profitability. This implies that **each lag interval likely requires its own separate optimization** routine.

Action Quality

The overarching goal for future iterations is to increase both metrics simultaneously.

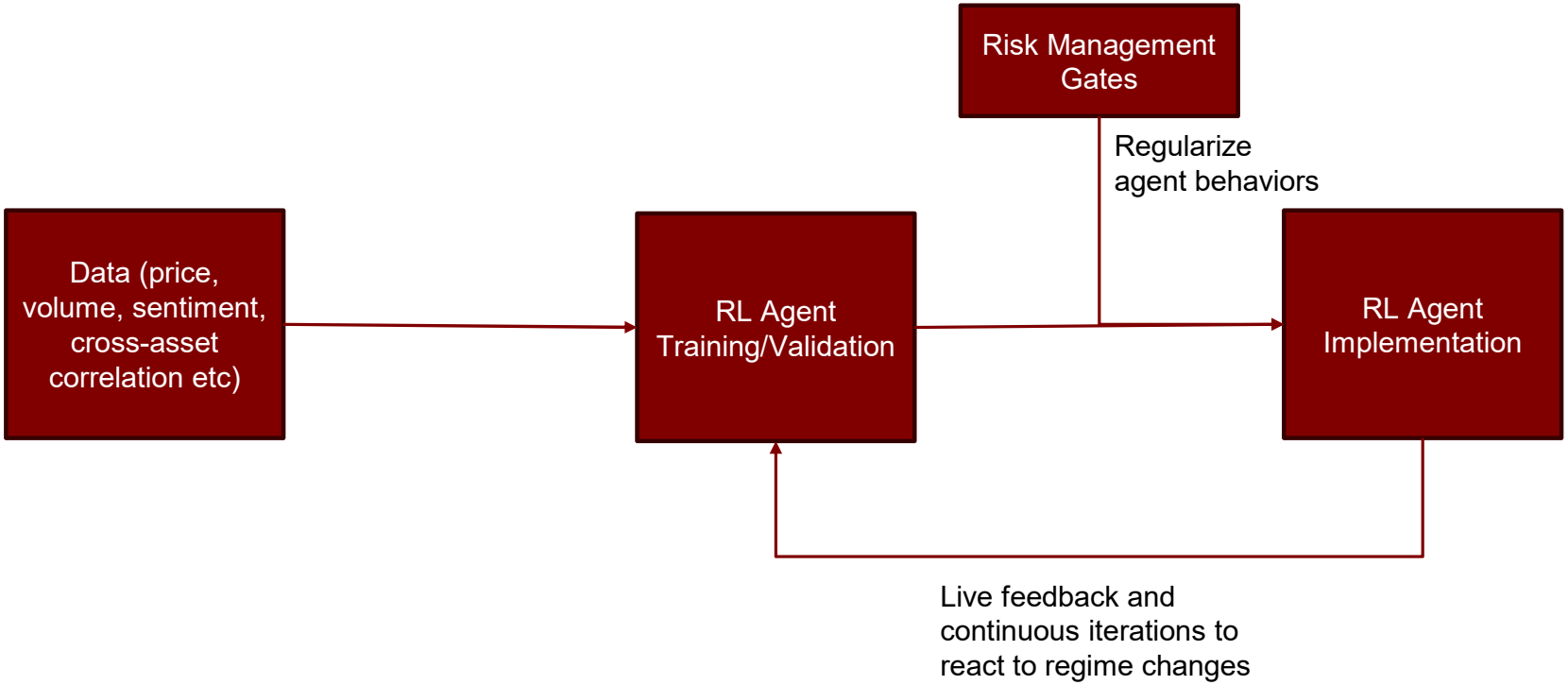
$$\text{Profit} = \text{Win Rate} \times \text{Avg. Profit}$$

Critical Insight

The agent may occasionally underperform during the in-sample training phase, yet outperform during out-of-sample testing.

Appendix

Appendix – Overall Workflow

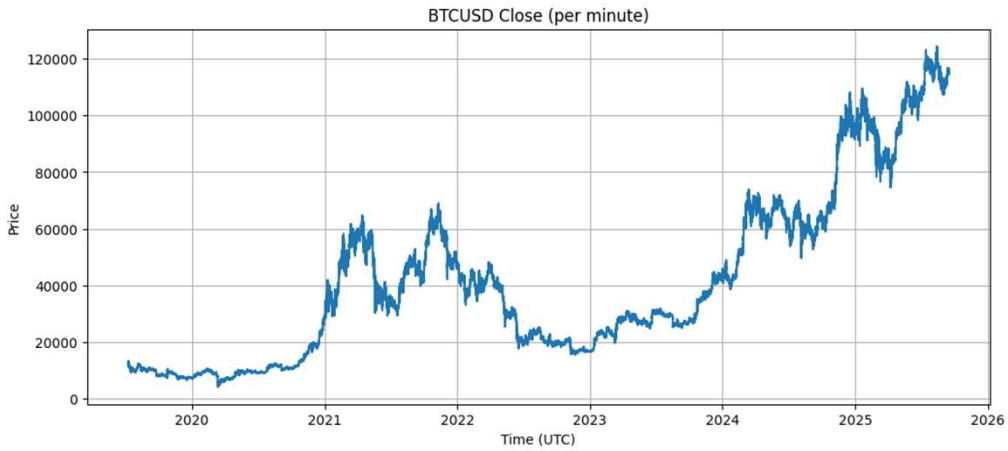


Data & Pipeline Overview

Raw Data

Loaded records: 3220762, columns: 8

	Date	Open	High	Low	Close	Tick Volume	Volume	Spread
0	2019.07.09 14:00	12561.65	12564.65	12560.65	12560.65	9	0	0
1	2019.07.09 14:01	12560.65	12560.65	12539.68	12552.66	43	0	0
2	2019.07.09 14:02	12552.66	12556.66	12545.67	12548.67	26	0	0
3	2019.07.09 14:03	12548.67	12548.67	12540.67	12544.67	9	0	0
4	2019.07.09 14:04	12544.67	12544.67	12544.67	12544.67	1	0	0



Cleaning & typing

- Parsed timestamps (`"%Y.%m.%d %H:%M"`) as UTC, then converted to local timezone (`TZ`)
- Coerced price/volume/spread columns to numeric; dropped rows with invalid timestamps
- Set `timestamp` as DatetimeIndex and sorted to ensure time-ordered series

Data quality checks

- **Per-minute completeness**: created an “expected” minute index from min→max timestamp and measured missing minutes
- **Gap detection**: grouped missing minutes into contiguous ranges (largest gaps summarized)
- **Duplicate timestamps**: checked duplicated minute timestamps (diagnostic printed)

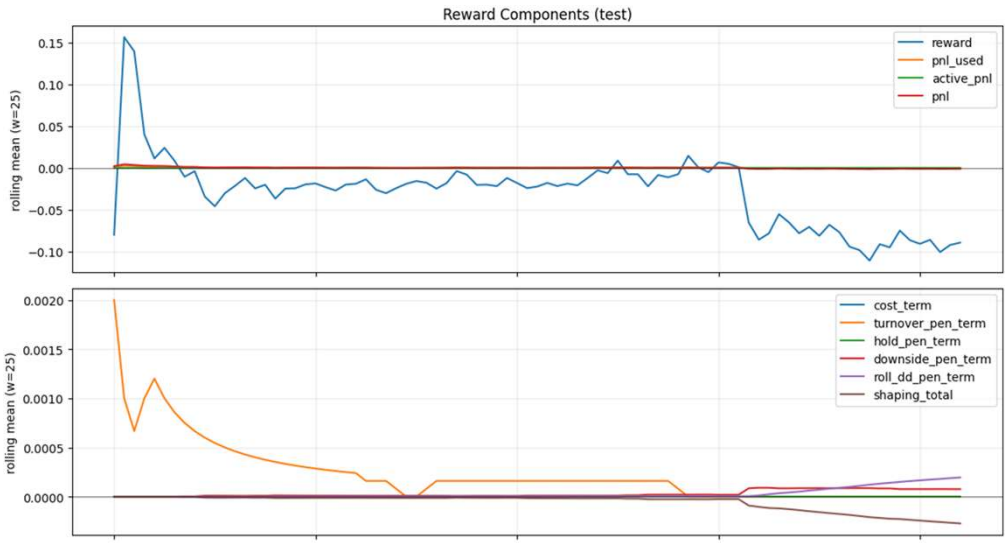
Threshold & Frequency Choice

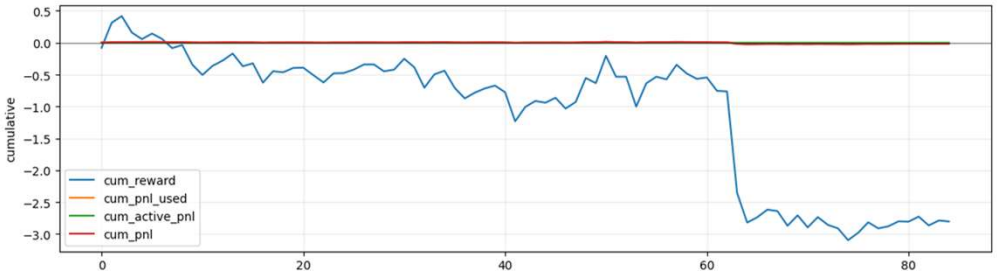
Threshold Selection:

- First, we explored both hourly and minute frequency, and noted that the hourly events have only **~8K**, too coarse for HFT decisions.
- We selected **0.25%** as the directional change threshold, as it retains **136K** observations for richer RL agent learning.
- Based on this threshold, over **50%** of the time, BTC continues to overshoot in the same direction **1.5x+** the original directional change movement, reflecting a sizable opportunity to monetize.

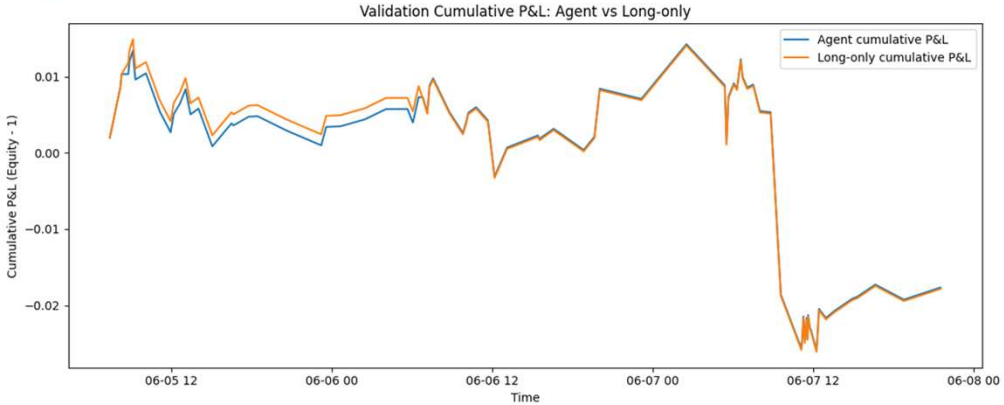
	dcc_move_ret	os_after_ret	os_ratio
count	136062.000000	136061.000000	136061.000000
mean	0.000017	0.000021	1.050179
std	0.003486	0.005575	1.391643
min	-0.050389	-0.148488	-0.000000
1%	-0.006238	-0.015629	0.000000
5%	-0.004198	-0.008071	0.000000
10%	-0.003573	-0.005347	0.000000
25%	-0.002909	-0.001898	0.108467
50%	0.000003	0.000000	0.616292
75%	0.002910	0.001920	1.460191
90%	0.003595	0.005401	2.634515
95%	0.004246	0.008236	3.584258
99%	0.006563	0.016079	6.288348
max	0.052681	0.146179	51.580960

Reward Function (Con't)






```
=== Test-period risk comparison (per exec interval; non-annualized) ===
Agent    Sharpe=-0.052943  Sortino=-0.061914  Calmar=-0.446032  MaxDD=0.039626  Std=0.003960
Long-only Sharpe=-0.053401  Sortino=-0.062477  Calmar=-0.441756  MaxDD=0.040416  Std=0.003967
Diff (Agent-Long): MaxDD=-0.000790  Std=-0.000007
```



We tested and monitored reward paths for multiple short episodes to fine tune hyperparameter selections, so agent has **balanced rewards and penalties**.

Regularization

Training

PPO:
- clip_range
- target_kl
- vf_coef
- entropy decay
- batch size / rollout sizing

Validation

RiskEval checkpoint filter:
- Score based on risk metrics.
- Guardrail reject for non-positive excess checkpoints.

Walk-forward candidate controls:
- Action-mix validity bounds (e.g. raw short/long action mix).
- Retry mechanism with alternate seeds when validation mix/regime signals look misaligned.
- Benchmark-aware validation score.
- Switch deployment to long-only when validation excess is below threshold.

Implementation

Action gates:
- Long-bias gate, confidence-hold gate, short-confidence gate.

Inference action-mix controller:
- Caps/floors for long/short/hold exposure; optional regime-adaptive limits.

Warm-start / normalization carry controls:
- Warm-start conditions and reset rules.
- VecNormalize carry with decay/blend to avoid stale-stat overfit.

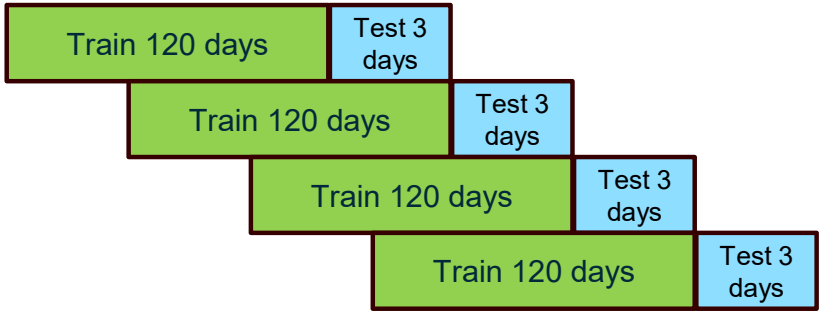
Execution safeguards:
- Optional stop-loss / take-profit liquidation with lagged execution price.

Training Protocol

One-time train/test (baseline)

Observations: slower to adapt to regime changes, due to mismatches of features' distribution between train/test

Walk-forward rolling split (High Frequency Trading style)



Observations: quicker to adapt to regime changes for lower frequency, but BTC price moves can still be volatile week over week (train/test mismatch); this more mimics a real-life trading model.

Training Protocol (WF)

As part of the walk-forward model, we evaluated correlation and sign agreement between various train/validation/test window lengths:

- train_days_grid=(120, 150, 180, 210, 240),
- val_days_grid=(7, 10, 14, 21, 28, 35, 42, 49, 56),
- test_days_grid=(3, 5, 7, 10, 14, 21, 28),

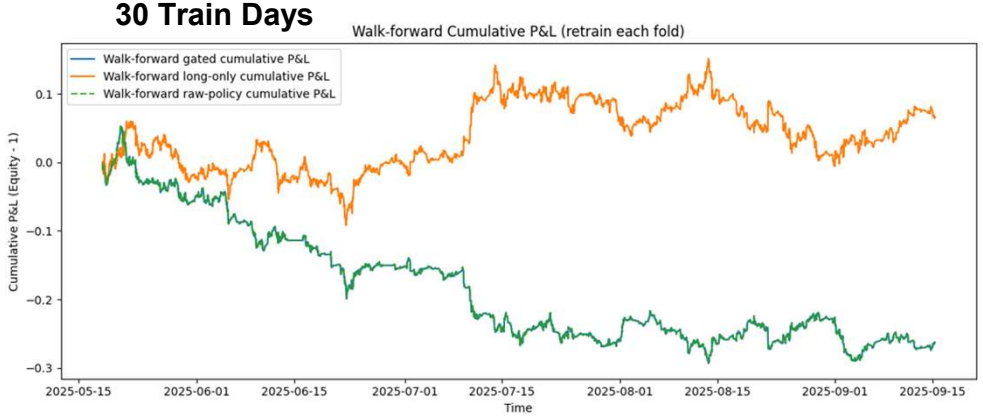
Observations:

- Shorter test windows captured near-term regime alignment more effectively
- Even for the top-ranked configurations, predictive alignment metrics remain modest: **Pearson is ~0.23, Spearman ~0.15, and sign agreement ~0.56**, indicating a limited directional edge.

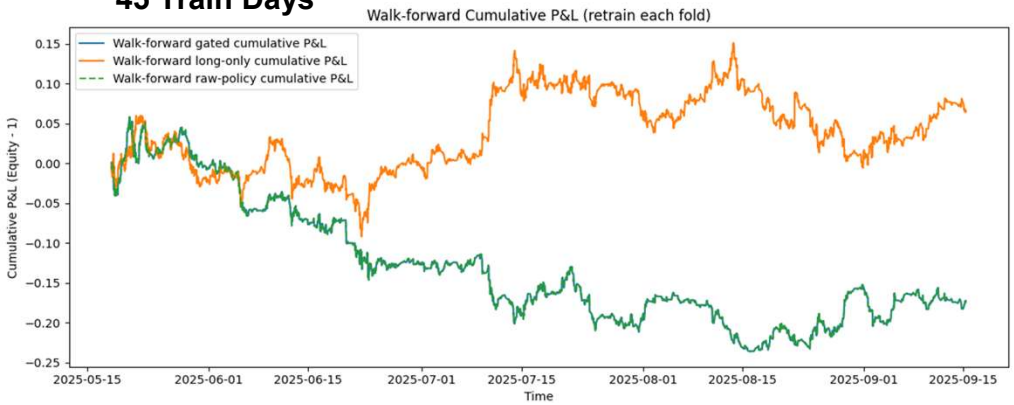
	train_days	val_days	test_days	step_days	n_folds	pearson	spearman	sign_agree	mean_val_ret	mean_test_ret	score
0	150	14	3	7	291	0.229253	0.154328	0.556701	0.023684	0.004736	0.370500
1	150	14	5	7	291	0.226397	0.155501	0.556701	0.023684	0.006661	0.370280
2	150	42	5	7	287	0.165333	0.158940	0.567944	0.079621	0.008437	0.364721
3	150	28	5	7	289	0.168747	0.148893	0.567474	0.049762	0.007081	0.362154
4	150	21	28	7	286	0.146739	0.156270	0.559441	0.037760	0.052736	0.355949
5	150	42	3	7	287	0.156795	0.150599	0.557491	0.079621	0.006204	0.355284
6	180	28	21	7	282	0.137158	0.147204	0.567376	0.056046	0.040754	0.355281
7	180	21	21	7	283	0.114828	0.158490	0.565371	0.041351	0.040796	0.353198
8	150	35	3	7	288	0.158404	0.156642	0.548611	0.064468	0.005056	0.352979
9	150	35	5	7	288	0.156617	0.152034	0.552083	0.064468	0.007284	0.352975
10	150	14	28	7	287	0.134922	0.146727	0.560976	0.024334	0.050979	0.351490
11	180	35	3	7	284	0.111386	0.144892	0.566901	0.070535	0.002389	0.349196
12	150	28	3	7	289	0.172275	0.147085	0.539792	0.049762	0.005178	0.348477
13	240	10	3	7	279	0.165978	0.133006	0.548387	0.018244	0.002865	0.347291
14	180	28	14	7	283	0.118760	0.152880	0.551237	0.055716	0.026726	0.345234
15	210	28	14	7	279	0.108664	0.145841	0.555556	0.056049	0.025066	0.343263
16	150	28	28	7	285	0.161282	0.153263	0.529825	0.051150	0.053909	0.343147
17	150	56	5	7	285	0.167170	0.119288	0.547368	0.113358	0.008601	0.342905
18	180	49	3	7	282	0.104802	0.114252	0.574468	0.103206	0.002462	0.342470
19	240	10	28	7	275	0.111012	0.119922	0.567273	0.018508	0.051462	0.341815

Evaluation (WF – different train days)

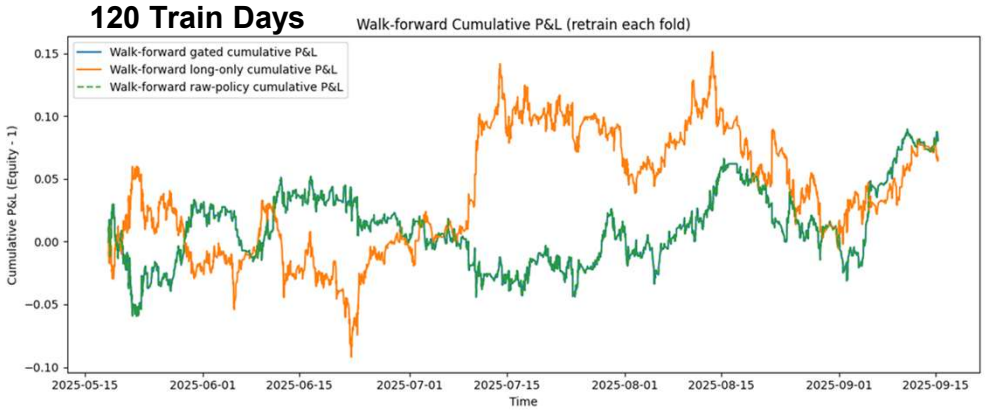
30 Train Days



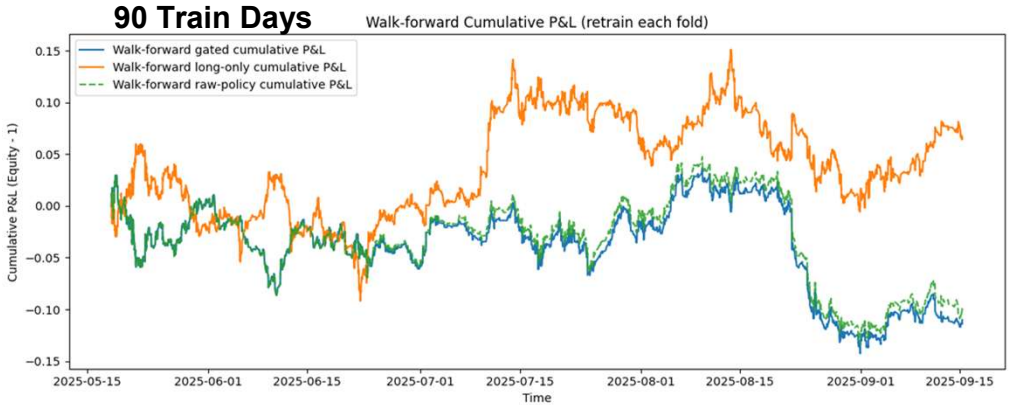
45 Train Days



120 Train Days

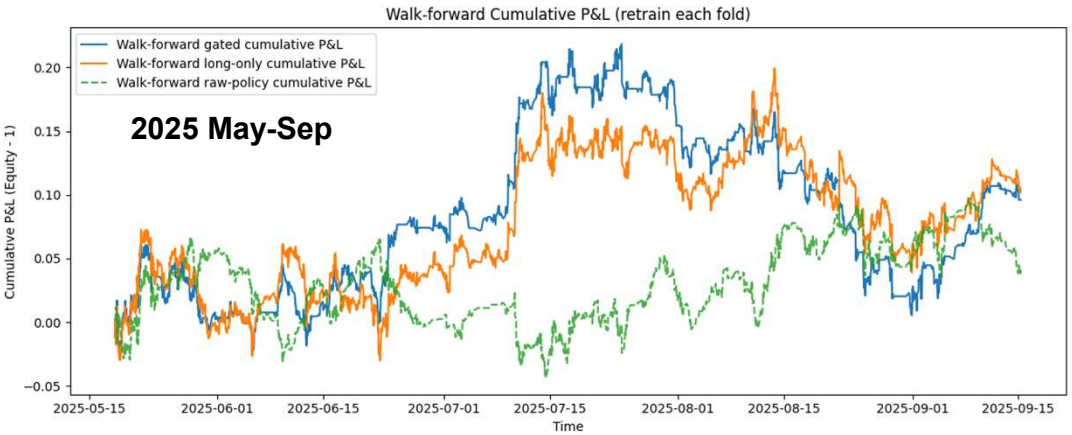


90 Train Days



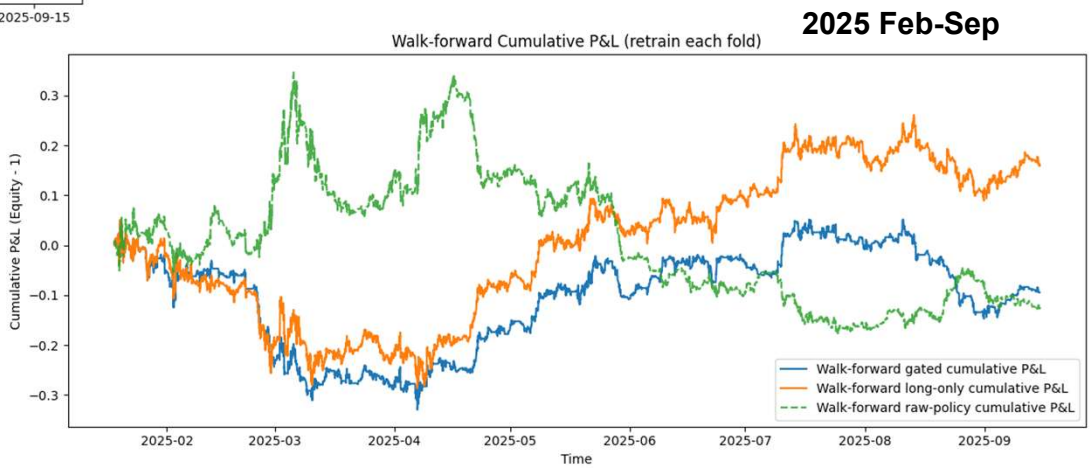
Note: validation days are 7, and test days are 5 days throughout these tests.

Evaluation (WF – different time windows)



Observations:

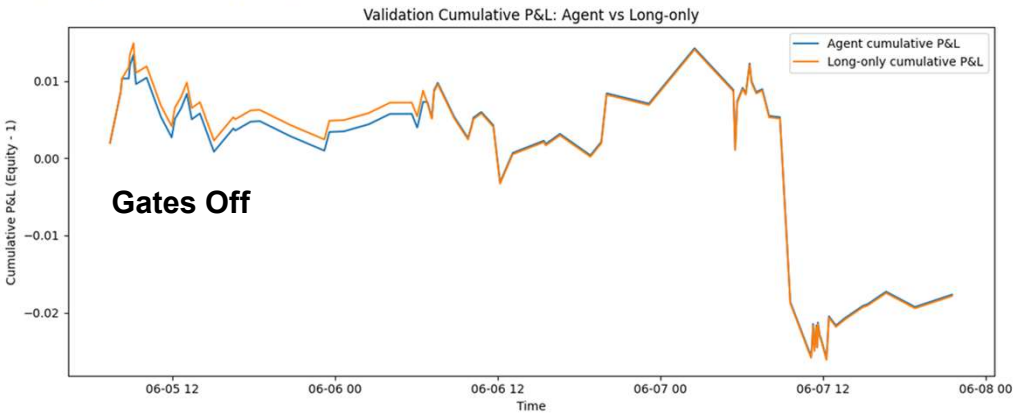
- Post-training regularization creates another layer of complexity, but is useful for risk control and mitigating action collapse (eg 100% short in an uptrend).
- Given the potential rapid regime change, agent could perform well during validation, but poorly in test if no regularization.



Note: train/validation/test days are 150/14/3, respectively, throughout these tests.

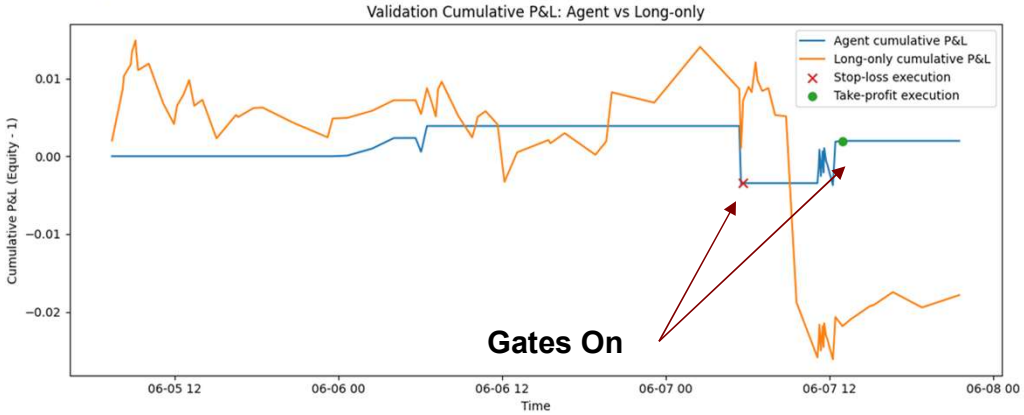
Evaluation (Profit taking/loss cutting)

```
=== Test-period risk comparison (per exec interval; non-annualized) ===
Agent    Sharpe=-0.052943  Sortino=-0.061914  Calmar=-0.446032  MaxDD=0.039626  Std=0.003960
Long-only Sharpe=-0.053401  Sortino=-0.062477  Calmar=-0.441756  MaxDD=0.040416  Std=0.003967
Diff (Agent-Long): MaxDD=-0.000790  Std=-0.000007
```



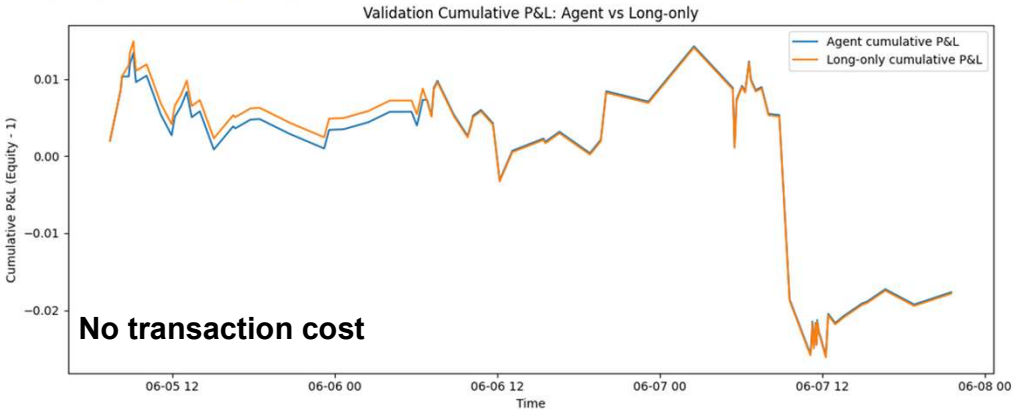
We explored profit-taking and loss-cutting gates to mimic a real-life trading environment for risk control. Similarly, these gates require further fine-tuning to produce risk-adjusted performance consistently.

```
=== Test-period risk comparison (per exec interval; non-annualized) ===
Agent    Sharpe=0.018616  Sortino=0.023963  Calmar=0.260647  MaxDD=0.007576  Std=0.001340
Long-only Sharpe=-0.053401  Sortino=-0.062477  Calmar=-0.441756  MaxDD=0.040416  Std=0.003967
Diff (Agent-Long): MaxDD=-0.032840  Std=-0.002627
```



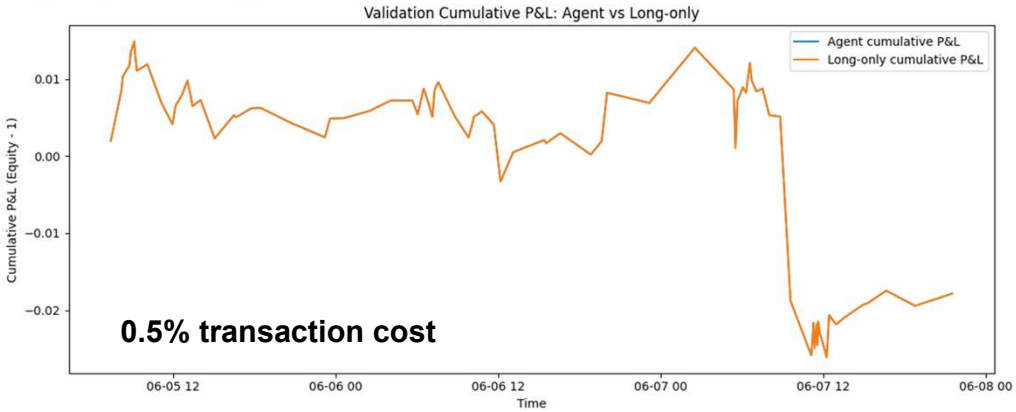
Evaluation (Transaction cost impact)

```
=== Test-period risk comparison (per exec interval; non-annualized) ===
Agent      Sharpe=-0.052943  Sortino=-0.061914  Calmar=-0.446032  MaxDD=0.039626  Std=0.003960
Long-only Sharpe=-0.053401  Sortino=-0.062477  Calmar=-0.441756  MaxDD=0.040416  Std=0.003967
Diff (Agent-Long): MaxDD=-0.000790  Std=-0.000007
```




```
=== Validation per-action ===
                n      freq  avg_reward  med_reward  win_rate  avg_pnl_frac_raw  \
action_name
HOLD            2  0.024096   -0.196537   -0.196537  0.000000          0.000000
LONG           81  0.975904   -0.029493    0.004097  0.530864         -0.000212
```

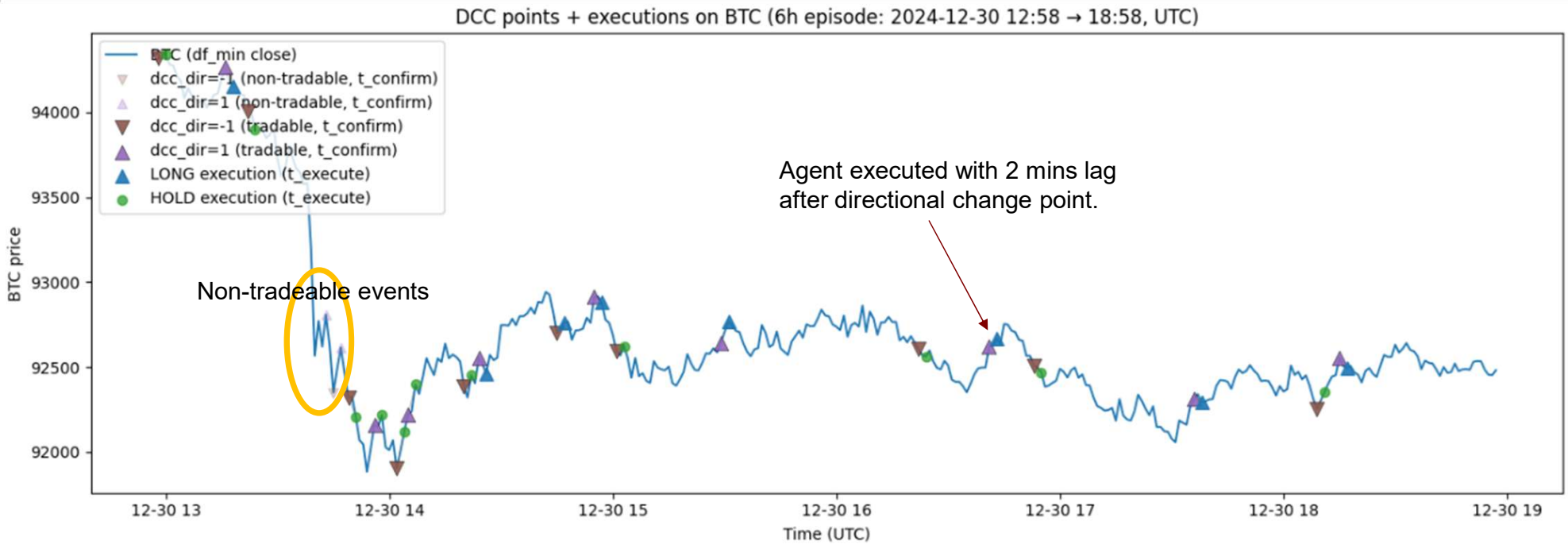
```
=== Test-period risk comparison (per exec interval; non-annualized) ===
Agent      Sharpe=-0.053401  Sortino=-0.062477  Calmar=-0.441756  MaxDD=0.040416  Std=0.003967
Long-only Sharpe=-0.053401  Sortino=-0.062477  Calmar=-0.441756  MaxDD=0.040416  Std=0.003967
Diff (Agent-Long): MaxDD=0.000000  Std=0.000000
```




```
=== Validation per-action ===
                n  freq  avg_reward  med_reward  win_rate  avg_pnl_frac_raw  \
action_name
LONG           83   1.0   -0.030222    0.022288  0.542169         -0.000209
```

Observations: we explored the impact of transaction cost, and noted agent trades less frequently. Currently, BTC transaction cost (on-chain directly) is below $1, less than 0.002%, so it may not be a key factor deterring agent behavior.

Evaluation (Random episode)



Evaluation – Overall action quality

```
=== Reward component diagnostics (group_col=action_exec_name, tradable_only=True, use_active_pnl=True) ===
                    n  avg_pnl_raw  avg_active_pnl_raw  avg_pnl_scaled  \
action_exec_name
HOLD             6079     0.000000           -0.000046       -0.004584
LONG             6271    -0.000012            0.000000        0.000000
SHORT             139    -0.000169           -0.000338       -0.033769

                 avg_cost  avg_hold_pen  avg_downside_pen  avg_reward_recon  \
action_exec_name
HOLD                  0.0      0.000592          0.002562         -0.007738
LONG                  0.0      0.000000          0.000000          0.000000
SHORT                 0.0      0.000000          0.008390         -0.042160

                 avg_reward_env  win_rate_pnl_used  win_rate_pnl_raw  \
action_exec_name
HOLD                  -0.007794           0.471788          0.000000
LONG                  -0.000056           0.000000          0.470579
SHORT                 -0.042240           0.539568          0.539568

                 win_rate_active_pnl
action_exec_name
HOLD                        0.471788
LONG                        0.000000
SHORT                       0.539568
```

Profit (action quality) = win rate * average profit

Observations: ideally case is with a high win rate and positive average profit. These are two key metrics we want to further improve, if time permits.

Appendix – Execution log sample

2 min lag

Reinvest all, or nothing

Agent live account balance

=== Execution log (head) ===

	k	idx_entry	idx_exit	decision_time	entry_time	exit_time	action	action_name	position	entry_price	exit_price	ret_interval	pnl_frac	equity_start	equity_end	pnl_abs
0	0	0	1	2024-11-20 21:37:00	2024-11-20 21:39:00	2024-11-20 21:56:00	2	LONG	1.0	97502.98	97008.38	-0.005073	-0.005073	1.000000	0.994927	-0.005073
1	1	1	2	2024-11-20 21:54:00	2024-11-20 21:56:00	2024-11-20 22:05:00	2	LONG	1.0	97008.38	97092.28	0.000865	0.000865	0.994927	0.995788	0.000860
2	2	2	3	2024-11-20 22:03:00	2024-11-20 22:05:00	2024-11-20 22:15:00	1	HOLD	0.0	97092.28	97146.48	0.000558	0.000000	0.995788	0.995788	0.000000
3	3	3	4	2024-11-20 22:13:00	2024-11-20 22:15:00	2024-11-20 22:19:00	2	LONG	1.0	97146.48	96938.68	-0.002139	-0.002139	0.995788	0.993658	-0.002130
4	4	4	5	2024-11-20 22:17:00	2024-11-20 22:19:00	2024-11-20 22:25:00	1	HOLD	0.0	96938.68	97337.78	0.004117	0.000000	0.993658	0.993658	0.000000
5	5	5	6	2024-11-20 22:23:00	2024-11-20 22:25:00	2024-11-20 22:49:00	2	LONG	1.0	97337.78	97137.68	-0.002056	-0.002056	0.993658	0.991615	-0.002043
6	6	6	7	2024-11-20 22:47:00	2024-11-20 22:49:00	2024-11-20 23:18:00	2	LONG	1.0	97137.68	97189.88	0.000537	0.000537	0.991615	0.992148	0.000533
7	7	7	8	2024-11-20 23:16:00	2024-11-20 23:18:00	2024-11-20 23:29:00	2	LONG	1.0	97189.88	97136.18	-0.000553	-0.000553	0.992148	0.991600	-0.000548
8	8	8	9	2024-11-20 23:27:00	2024-11-20 23:29:00	2024-11-21 00:22:00	2	LONG	1.0	97136.18	97007.68	-0.001323	-0.001323	0.991600	0.990288	-0.001312
9	9	9	10	2024-11-21 00:20:00	2024-11-21 00:22:00	2024-11-21 00:28:00	2	LONG	1.0	97007.68	96872.98	-0.001389	-0.001389	0.990288	0.988913	-0.001375
10	10	10	11	2024-11-21 00:26:00	2024-11-21 00:28:00	2024-11-21 00:54:00	2	LONG	1.0	96872.98	97042.98	0.001755	0.001755	0.988913	0.990648	0.001735
11	11	11	12	2024-11-21 00:52:00	2024-11-21 00:54:00	2024-11-21 02:12:00	2	LONG	1.0	97042.98	97455.08	0.004247	0.004247	0.990648	0.994855	0.004207
12	12	12	13	2024-11-21 02:10:00	2024-11-21 02:12:00	2024-11-21 02:30:00	2	LONG	1.0	97455.08	97625.48	0.001748	0.001748	0.994855	0.996595	0.001740
13	13	13	14	2024-11-21 02:28:00	2024-11-21 02:30:00	2024-11-21 03:14:00	2	LONG	1.0	97625.48	98241.48	0.006310	0.006310	0.996595	1.002883	0.006288
14	14	14	15	2024-11-21 03:12:00	2024-11-21 03:14:00	2024-11-21 04:04:00	2	LONG	1.0	98241.48	98079.88	-0.001645	-0.001645	1.002883	1.001233	-0.001650
15	15	15	16	2024-11-21 04:02:00	2024-11-21 04:04:00	2024-11-21 04:14:00	2	LONG	1.0	98079.88	97883.78	-0.001999	-0.001999	1.001233	0.999232	-0.002002
16	16	16	17	2024-11-21 04:12:00	2024-11-21 04:14:00	2024-11-21 04:23:00	2	LONG	1.0	97883.78	98161.88	0.002841	0.002841	0.999232	1.002070	0.002839
17	17	17	18	2024-11-21 04:21:00	2024-11-21 04:23:00	2024-11-21 04:33:00	2	LONG	1.0	98161.88	97825.48	-0.003427	-0.003427	1.002070	0.998636	-0.003434
18	18	18	19	2024-11-21 04:31:00	2024-11-21 04:33:00	2024-11-21 04:42:00	1	HOLD	0.0	97825.48	97960.88	0.001384	0.000000	0.998636	0.998636	0.000000
19	19	19	20	2024-11-21 04:40:00	2024-11-21 04:42:00	2024-11-21 04:51:00	2	LONG	1.0	97960.88	97648.78	-0.003186	-0.003186	0.998636	0.995455	-0.003182
20	20	20	21	2024-11-21 04:49:00	2024-11-21 04:51:00	2024-11-21 05:03:00	2	LONG	1.0	97648.78	97541.58	-0.001098	-0.001098	0.995455	0.994362	-0.001093
21	21	21	22	2024-11-21 05:01:00	2024-11-21 05:03:00	2024-11-21 05:22:00	2	LONG	1.0	97541.58	97365.38	-0.001806	-0.001806	0.994362	0.992566	-0.001796
22	22	22	23	2024-11-21 05:20:00	2024-11-21 05:22:00	2024-11-21 05:34:00	1	HOLD	0.0	97365.38	97530.18	0.001693	0.000000	0.992566	0.992566	0.000000
23	23	23	24	2024-11-21 05:32:00	2024-11-21 05:34:00	2024-11-21 06:03:00	2	LONG	1.0	97530.18	97442.48	-0.000899	-0.000899	0.992566	0.991673	-0.000893
24	24	24	25	2024-11-21 06:01:00	2024-11-21 06:03:00	2024-11-21 06:20:00	2	LONG	1.0	97442.48	97499.58	0.000586	0.000586	0.991673	0.992254	0.000581

Appendix – Action gate sample

After agent assesses action probabilities at each decision-making point…

Long \ Hold	5%	10%	15%	20%	25%	30%	35%	40%	45%	50%	55%	60%	65%	70%	75%	80%	85%	90%	95%
5%	90%	85%	80%	75%	70%	65%	60%	55%	50%	45%	40%	35%	30%	25%	20%	15%	10%	5%	0%
10%	85%	80%	75%	70%	65%	60%	55%	50%	45%	40%	35%	30%	25%	20%	15%	10%	5%		
15%	80%	75%	70%	65%	60%	55%	50%	45%	40%	35%	30%	25%	20%	15%	10%	5%			
20%	75%	70%	65%	60%	55%	50%	45%	40%	35%	30%	25%	20%	15%	10%	5%				
25%	70%	65%	60%	55%	50%	45%	40%	35%	30%	25%	20%	15%	10%	5%					
30%	65%	60%	55%	50%	45%	40%	35%	30%	25%	20%	15%	10%	5%						
35%	60%	55%	50%	45%	40%	35%	30%	25%	20%	15%	10%	5%							
40%	55%	50%	45%	40%	35%	30%	25%	20%	15%	10%	5%	0%							
45%	50%	45%	40%	35%	30%	25%	20%	15%	10%	5%	0%								
50%	45%	40%	35%	30%	25%	20%	15%	10%	5%	0%									
55%	40%	35%	30%	25%	20%	15%	10%	5%	0%										
60%	35%	30%	25%	20%	15%	10%	5%	0%											
65%	30%	25%	20%	15%	10%	5%													
70%	25%	20%	15%	10%	5%														
75%	20%	15%	10%	5%															
80%	15%	10%	5%																
85%	10%	5%																	
90%	5%																		
95%																			

Agent is only allowed to SHORT when short probability assessment is above 75%.

Forcing agent to HOLD, if hold probability assessment is above 75%.

To explore: the accuracy of probability assessment is an area for further refinement, which may require more features and exploration beyond just the PPO model.